

Huayu Chen

📍 Beijing, China ✉ chenhuay23@mails.tsinghua.edu.cn ☎ 15071064154
🔗 <https://huayuchen2004.github.io/> 🌐 HuayuChen2004

Research Interest

My research interest lies in computer vision, reinforcement learning and causal perception, and have been reading related papers for the past year. I love to learn new knowledge, to think and to write. My ultimate dream is to build a human-like machine utilizing the power of computation and memory, and this requires extending robots' abilities in various fields, such as perception, motion, cognition and reasoning. I will work ceaselessly to approach this goal.

Education

Tsinghua University

Sept 2024 – now

Undergrad in Department of Automation

- GPA: 3.77/4.0
- Maths: Calculus, Linear Algebra, Discrete Mathematics, Basic Probability and Statistics
- Computer Science: Programming in C, Object Oriented Programming, Data Structure, Introduction to AI
- Electronic Engineering: Electric Circuit, Digital Electronic Technology

Experience

Beijing Institute for General Artificial Intelligence

July 2024 – now

- Took courses in various fields in AI, read extensive papers, wrote a reading report and a research essay in the summer practical program
- Volunteered in two projects working separately on 3D reconstruction and decision-making process

Social Work

Association of Science and Technology of Automation, Tsinghua University

July 2024 – now

- Received extensive training on basic programming skills and team development
- Took part in the development of a programming competition Thuai-8

Projects

Cooperative Project On Decision Making and Temporal Abstraction

Mar 2024 – now

- Took part in the development of experimental environment
- Ran extensive experiments to test offline and online RL and DT algorithms

Cooperative Project On 3D Reconstruction

Oct 2024 – now

- Processed video data
- Worked on multi-state modeling with monocular video input
- Helped improve learning algorithms to promote efficient and accurate learning

Style Transfer

[github](#) 

- Made use of large vision model to take features from images and transfer to other images
- Automated large scale training process

Street Cleaning

[github](#) 

- Developed a training environment that is compatible with OpenAI gym APIs to test RL algorithms
- Tested both traditional planning algorithms and deep reinforcement learning algorithms to try to solve the problem of street cleaning
- Visualized results using pygame

Nonogram

[github](#) 

- Developed Nonogram game using PyQt as visualization
- Tested some algorithms to solve Nonogram with less computing complexity

3DWorldBuilding

[github](#) 

- Project for Object Oriented Programming
- Support operations on points, lines, and faces in obj files that can be visualized online

Skills

Code Languages: C++, C, Python, C#, MATLAB

Technologies: Linux basic commands, Git multi-branch development, Research tools such as tmux and wandb, Neural networks with pytorch, Pygame and PyQt GUI